

The Linear Model

OLS, Inference, and Diagnostics

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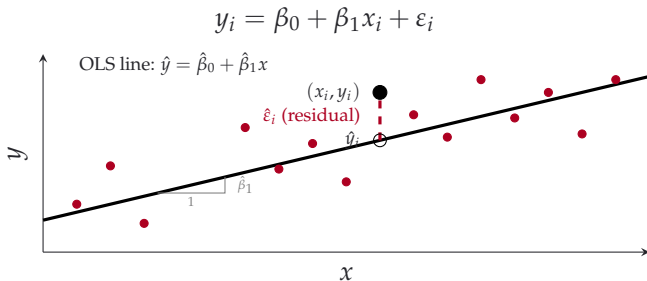
M1 ECO

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Where OLS comes from: Gauss and Ceres

- **1801** — Giuseppe Piazzi spots **Ceres** (a dwarf planet in the asteroid belt), then loses it behind the Sun after only a handful of noisy observations.
- **The question:** *where will it reappear?*
- The young **Carl Friedrich Gauss** computes the orbit with a new method — choose the trajectory that **minimises the sum of squared residuals** (the gaps between the observed positions and the predicted ones).
- **Result:** an accurate prediction; Ceres is found again. **Ordinary Least Squares (OLS)** is born.

OLS in one picture



OLS picks $(\hat{\beta}_0, \hat{\beta}_1)$ by **least squares** — minimising:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{\beta_0, \beta_1} \sum_i \underbrace{(y_i - \beta_0 - \beta_1 x_i)^2}_{\text{sum of squared residuals}}.$$

ε_i

Solving for $\hat{\beta}$ (simple regression)

Minimising the sum of squared residuals $S(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$, set its two derivatives to zero:

$$\sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0, \quad \sum_i x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0.$$

- The first gives $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ — the line passes through $(\bar{x}, \bar{y})$.
- Substituting into the second and solving for the slope:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)}.$$

($\bar{x} = \frac{1}{n} \sum_i x_i$ is the sample mean; a hat marks an estimate. The $1/n$ in Cov and Var cancels, so this ratio of moments equals the ratio of sums above.)

Multiple regression

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \varepsilon_i$$

- Here k is the **number of explanatory variables** (slopes); with the intercept there are $k + 1$ coefficients.
- Each β_j is the effect of x_j **holding the other regressors fixed** — identified from the variation in x_j *conditional on* the others.
- OLS minimises $\sum_i (y_i - x_i' \beta)^2$, with closed-form solution

$$\hat{\beta} = (X'X)^{-1}X'y$$

where X is the $n \times (k+1)$ matrix of regressors and a **prime** denotes the **transpose** (so X' is X flipped — rows become columns).

OLS by hand: a worked example

A small case — **3 observations, 2 parameters** — you can check on the board.
Model $y = \beta_0 + \beta_1 x$ with points $(x, y) = (1, 2), (2, 2), (3, 5)$:

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad y = \begin{pmatrix} 2 \\ 2 \\ 5 \end{pmatrix}, \quad X'X = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}, \quad X'y = \begin{pmatrix} 9 \\ 21 \end{pmatrix}.$$

$$(X'X)^{-1} = \frac{1}{6} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}, \quad \hat{\beta} = (X'X)^{-1}X'y = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1.5 \end{pmatrix}.$$

- First column of X is **1s** (the intercept); $\det(X'X) = 42 - 36 = 6$.
- **Check:** $\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{3}{2} = 1.5$ — the matrix formula reproduces the simple-regression slope. (More regressors $\Rightarrow X$ gains columns; need $n \geq k + 1$.)

Roadmap

We can already *compute* $\hat{\beta}$. The rest of the lecture asks four questions:

- 1 **Interpret** — what does a coefficient *mean*?
- 2 **Inference** — is it *real* (precise, significant)?
- 3 **Assumptions — and when they fail** — is it *valid* (unbiased, honest SE)?
- 4 **Doing it in Python** (pandas + statsmodels).

Section 1

Interpreting coefficients

Interpretation of coefficients

model	equation	a change in x ...	... changes y by
level-level	$y = \beta_0 + \beta_1 x$	+1 unit	β_1 units
log-level	$\ln y = \beta_0 + \beta_1 x$	+1 unit	$100 (e^{\beta_1} - 1) \%$
level-log	$y = \beta_0 + \beta_1 \ln x$	+1%	$\beta_1 / 100$ units
log-log	$\ln y = \beta_0 + \beta_1 \ln x$	+1%	$\beta_1 \%$ (elasticity)

For small β_1 these collapse to the familiar approximations (e.g. log-level $\approx 100\beta_1\%$); the exact forms matter only when β_1 is sizeable.

Example: the gender wage gap

$$\ln(\text{wage}_i) = \beta_0 + \beta_1 \text{woman}_i + \varepsilon_i, \quad \hat{\beta}_1 \approx -0.285.$$

Women earn on average $\approx 24.8\%$ less than men (INSEE, 2017). For a log outcome with a **dummy** regressor, use the exact effect:

$$e^{\hat{\beta}_1} - 1 = e^{-0.285} - 1 = -0.248 = -24.8\%.$$

Ceteris paribus

Add controls — education, experience, hours, occupation, sector:

$$\ln(\text{wage}_i) = \beta_0 + \beta_1 \text{woman}_i + \beta_2 \text{educ}_i + \cdots + \varepsilon_i, \quad \hat{\beta}_1 \approx -0.10.$$

- “**All else equal**,” women earn $\approx 9.5\%$ less — exactly $e^{-0.10} - 1 = -0.095$. The gap shrinks once we hold **observables** fixed, but only the *observed* ones.
- **Careful — not every control helps.** A variable that is itself a *consequence* of being a woman (occupation, sector — if discrimination channels women into lower-paid roles) is a **bad control**: adding it can *hide* part of the true gap. Which controls to include is already a causal question.

What's left to check

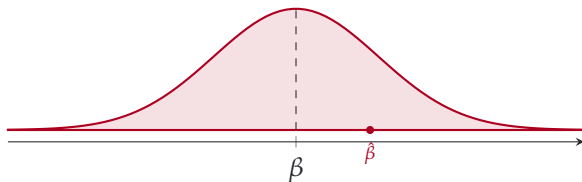
We have read these coefficients at face value. Two questions remain: are they statistically distinguishable from zero (**inference**, next), and are they unbiased in the first place (**assumptions**, after)?

Section 2

Inference

The true β we never observe

- Behind the data is a **population** (a data-generating process) with a **true but unknown** coefficient β — the effect of x on y for *everyone*. We never see it.
- From **one sample** of n units we compute an **estimate** $\hat{\beta}$. A *different* sample would give a *different* $\hat{\beta}$ — so $\hat{\beta}$ is **random**, with a **sampling distribution**:



One sample gives one draw $\hat{\beta}$ from this curve. **Inference** asks how far it is likely to land from the true β . (The curve is drawn centred at β — that OLS is unbiased is the next slide.)

How precise is $\hat{\beta}$? The standard error

- OLS is **unbiased** (and **consistent**): across samples $\hat{\beta}$ is centred on the truth, $\mathbb{E}(\hat{\beta}) = \beta$.
- The **standard error** is the **standard-deviation spread** of that curve, $\widehat{\text{se}}(\hat{\beta}) = \sqrt{\widehat{\text{Var}}(\hat{\beta})}$ — how far a typical $\hat{\beta}$ lands from β .
- In simple regression $\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}$, with $\sigma^2 = \text{Var}(\varepsilon_i)$ the error variance (estimated by $\hat{\sigma}^2 = \text{RSS}/(n - k - 1)$, with RSS the **residual sum of squares** $\sum_i \hat{\varepsilon}_i^2$). So the SE is **smaller** when n is large, the regressor varies more ($\sum_i (x_i - \bar{x})^2 = n \text{Var}(x)$), or residuals are tight.

$n - k - 1$ is the **degrees of freedom** — we “use up” $k + 1$ observations estimating the coefficients. This is the *classical* SE (valid under homoskedasticity); we will see it can be wrong, and how to fix it (robust SE).

Student's t : a story

- W. S. **Gosset**, a chemist at **Guinness**, needed to compare beer batches from **small samples** — and published under the pen-name “*Student*”.
- His **t -distribution** describes the estimator's behaviour when σ^2 is itself estimated and n is small — exactly our situation.
- *Hold your beer*: it is the workhorse of significance testing.

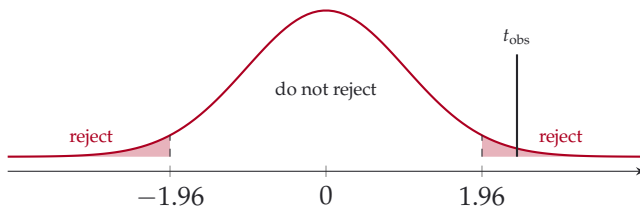
The t -statistic: $\hat{\beta}$ in standard-error units

To test $H_0 : \beta_k = \beta_k^0$ (usually $\beta_k^0 = 0$): how far is $\hat{\beta}_k$ from the hypothesised value, **measured in standard errors**?

$$t = \frac{\hat{\beta}_k - \beta_k^0}{\widehat{\text{se}}(\hat{\beta}_k)} \xrightarrow{H_0: \beta_k=0} \frac{\hat{\beta}_k}{\widehat{\text{se}}(\hat{\beta}_k)}.$$

- So t is the estimate in **se units**: $t = 2$ means $\hat{\beta}_k$ sits two standard errors from 0. A **large** $|t| \Rightarrow \hat{\beta}_k$ is far from 0 relative to its noise $\Rightarrow$ implausible if the true β_k were 0.
- Under the assumptions (normal errors), t follows **Student's t with $n - k - 1$ degrees of freedom**, which $\rightarrow \mathcal{N}(0, 1)$ as n grows.
- **Why t , not the normal?** Because we *estimate* σ^2 inside $\widehat{\text{se}}$; that extra uncertainty **fattens the tails** in small samples — Gosset's discovery.

Reading the t : critical values and rejection



- Under H_0 , t is centred at 0. **Reject** H_0 when t lands in the **tails**, beyond the **critical value**.
- Two-sided 5%: reject if $|t| > 1.96$ (shaded 2.5% + 2.5%). Other levels: 10% $\rightarrow$ 1.645, 1% $\rightarrow$ 2.58. (*Small n : the threshold is larger — e.g. ≈ 2.2 at 10 d.f.; read it off the t -table.*)
- **Rule of thumb:** $|\hat{\beta}_k| > 2 \hat{se} \Rightarrow$ significant at 5%. (The p -value, next, is the tail area beyond the observed $|t|$.)

p -values and confidence intervals

What a p -value is — and is not

The p -value is the probability, if H_0 **were true**, of a $|t|$ at least as large as the one observed. A small p means the data are unlikely under H_0 , so we reject.

It is **not** the probability that H_0 is true.

- **95% confidence interval:** $\hat{\beta}_k \pm 1.96 \widehat{\text{se}}(\hat{\beta}_k)$. If it **excludes 0**, reject H_0 at 5% (equivalent to the t -test).
- **But significance is not effect size**, and insignificance is not “no effect” — see the next slide.

Significance is not effect size: power and magnitude

Power — the chance of catching a true effect

The **power** of a test is the probability it rejects H_0 *when H_0 is false* — i.e. detects a real effect. It **ris**es with the sample size n , the size of the true effect, and precision (small SE).

- **Small $n \Rightarrow$ low power:** a real, theory-backed effect can look **insignificant** only because the SE is large. *Absence of evidence is not evidence of absence* — get **more data**, don't drop the variable.
- **Large $n \Rightarrow$ almost everything rejects H_0 :** even economically trivial effects become “significant.”

Judge the magnitude concretely: convert $\hat{\beta}$ to real units (via the functional form), compare it to a **benchmark** (the mean or SD of y , a policy threshold, other studies), and read the **whole CI** — a wide interval is *uninformative*, not *zero*.

Reading the whole output: R^2 and the F -test

R^2 — goodness of fit

Total variation in y splits into explained and residual parts:

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{\varepsilon}_i^2,$$

the *total* (TSS), *explained* (ESS) and *residual* (RSS) sum of squares. Then

$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} \in [0, 1]$ — the share of y 's variance the model explains.

- **High $R^2 \neq$ good estimate:** a biased $\hat{\beta}$ can fit well, and a valid causal effect can have low R^2 . *Fit $\neq$ correctness* (adjusted R^2 penalises extra regressors).
- **The F -test** checks **joint** significance (H_0 : several slopes = 0 at once) — also printed in every output. Handy under multicollinearity: individual t 's look insignificant, yet the joint F is significant.

Section 3

The assumptions — and when they fail

The OLS assumptions (Gauss–Markov)

- 1 **Linearity in parameters:** $y = X\beta + \varepsilon$.
- 2 **Exogeneity (zero conditional mean):** $\mathbb{E}(\varepsilon_i | X) = 0$.
- 3 **Homoskedasticity:** $\text{Var}(\varepsilon_i | X) = \sigma^2$ (constant).
- 4 **No autocorrelation:** $\text{Cov}(\varepsilon_i, \varepsilon_j | X) = 0, i \neq j$.
- 5 **No perfect multicollinearity** — needed for $(X'X)^{-1}$ to exist.

(Normality of the errors is added *only* for exact small-sample t -tests.) Here $\mathbb{E}[\cdot | X]$ is the **expectation** given the regressors; Cov, Var are covariance and variance.

What each assumption buys

1–2 $\Rightarrow$ OLS is **unbiased & consistent**. Adding 3–4 $\Rightarrow$ it is **BLUE** (minimum variance among linear unbiased estimators). Each "when it fails" slide drops one and asks what breaks.

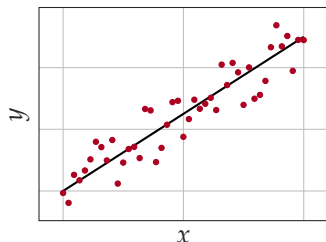
Heteroskedasticity: a variance problem

Heteroskedasticity is always a problem with the **variance (volatility)** of the forecast errors:

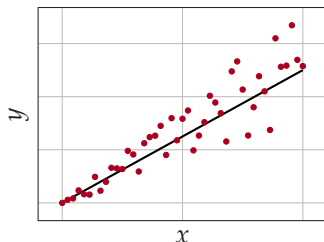
- **Cross-section:** the error variance varies with the explanatory variable x_i .
- **Time series:** the error variance varies with the explanatory variable x_t .
- **Spatial (cross-sectional) form:** the error *variance* differs systematically across places — e.g. larger in some regions than others. (*Variance, not correlation — error correlation across units is autocorrelation, next.*)
- **Another time-series form:** the error variance at t varies with the variance of *past* errors.

Homoskedasticity vs. heteroskedasticity

Homoskedastic



Heteroskedastic



Left: residual spread is roughly **constant**. Right: it **fans out** with x . Beware — correlated errors can look similar, so rely on **formal tests**, not the eye.

Heteroskedasticity: detection

$\text{Var}(\varepsilon_i \mid X)$ is **not** constant. Then OLS is still **unbiased** but no longer efficient, and **default standard errors are wrong** $\Rightarrow$ invalid t -tests.

Tests (H_0 : homoskedasticity)

Breusch–Pagan: regress $\hat{\varepsilon}_i^2$ on the regressors, $\hat{\varepsilon}_i^2 = \delta_0 + \delta_1 x_{1i} + \cdots + v_i$, and test $\delta_1 = \cdots = 0$.

White: same idea, adding *squares* and *interactions*.

$p < 0.05 \Rightarrow$ reject homoskedasticity. ($\hat{\varepsilon}_i = y_i - x_i' \hat{\beta}$ are the **residuals** — the estimated errors.)

Heteroskedasticity: the fix

Use robust standard errors by default

Heteroskedasticity-consistent (**HC** / White) standard errors fix inference without changing $\hat{\beta}$.

- The **sandwich** covariance (White, 1980):

$$\widehat{\text{Var}}(\hat{\beta}) = (X'X)^{-1} X' \text{diag}(\hat{\varepsilon}_i^2) X (X'X)^{-1},$$

where $\text{diag}(\hat{\varepsilon}_i^2)$ is the diagonal matrix of squared residuals.

- $\hat{\beta}$ is **unchanged** — only the SE (hence t , p , CIs) are corrected.
- The form shown is **HC0**; finite-sample versions inflate the squared residuals — **HC3** is the recommended default.
- Alternative: **GLS** / **FGLS** model the variance explicitly (less common in practice).

Autocorrelation: an error-correlation problem

Autocorrelation is no longer about the *variance* of the error, but about the **error itself being correlated** across observations:

- **Cross-section:** one unit's errors affect another's — when units share common characteristics or factors that create dependence. *Example: spatial data.*
- **Time series:** the error at t is a function of past errors ($t - 1, t - 2, \dots$).

Autocorrelation: tests and fix

Formally, $\text{Cov}(\varepsilon_i, \varepsilon_j) \neq 0$:

- **spatial:** $\varepsilon_i = \rho \sum_{j \neq i} w_{ij} \varepsilon_j + u_i$ (w_{ij} a proximity weight);
- **time series:** $\varepsilon_t = \rho_1 \varepsilon_{t-1} + \dots + \rho_p \varepsilon_{t-p} + u_t$.

OLS stays unbiased, but **standard errors are wrong**.

- **Tests:** Durbin–Watson ($DW \approx 2$: none; ≈ 0 : positive; ≈ 4 : negative — but invalid with a lagged dependent variable); Breusch–Godfrey (higher-order, valid with lagged y).
- **Fix:** HAC SE (Newey–West, 1987) — robust to *both* heteroskedasticity and autocorrelation; or Cochrane–Orcutt / Prais–Winsten.

Multicollinearity

Two or more regressors are strongly linearly related, e.g. $x_1 \approx a + b x_2$:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon.$$

- Consequences: **unstable** coefficients (signs flip), **inflated** standard errors / wide CIs, hard to separate individual effects. ($\hat{\beta}$ *stays unbiased*.)
- **Detect:** variance inflation factor (VIF). **Fix:** drop or combine variables; get more data.
- *Example:* regressing salary on **age and experience** — highly correlated, so β_1, β_2 are hard to pin down.

Non-normality of residuals

- Normal errors are needed for **exact small-sample** t -tests — **not** for unbiasedness, and not asymptotically (the CLT rescues large n).
- **Test:** Jarque–Bera (H_0 : residuals are normal); $p < 0.05 \Rightarrow$ reject.
- If violated **and** n small: use the **bootstrap** — resample the data with replacement many times to approximate the sampling distribution of $\hat{\beta}$.

Endogeneity: the deepest problem

When $\mathbb{E}(\varepsilon \mid X) \neq 0$, OLS is **biased** — and no robust SE can fix that. Three sources:

- **omitted variables, reverse causality, measurement error.**

Which way does omitted-variable bias go? Its sign is (effect of the omitted variable on y) $\times$ (its correlation with x). *Wage gap*: education/ability raise wages and correlate with the woman dummy in the raw data, biasing the uncontrolled gap — adding controls moved $-0.285 \rightarrow -0.10$.

Causal inference (the rest of the course)

Designed to defeat the first two. Four workhorses: randomized experiments, **difference-in-differences**, regression discontinuity, and **instrumental variables** (later: shift-share).

Section 4

Estimating in Python

The toolkit: which package, for what

package	use it for
pandas	load, inspect, clean, merge, transform data (the DataFrame)
numpy	numerical operations (logs, powers, arrays)
statsmodels	estimation + inference (coef, se, t , p , CI) and diagnostic tests
matplotlib	plots and visual checks

Typical workflow: **load** → **inspect** → **transform** → **estimate** → **robust SE** → **diagnostics**. (Fixed effects and IV come later, with pyfixest.)

Load and inspect (pandas)

```
import pandas as pd

df = pd.read_stata("wage.dta")          # or read_csv / read_excel
df.head(); df.describe(); df.info()    # inspect first!
df = pd.merge(df, dr, on="id", how="inner")
```

- **Always inspect before estimating:** types, ranges, missing values, oddities.
- merge: inner keeps matched rows, outer keeps all (gaps → NaN), left/right keep one side. Windows paths: prefix with r"...".

Transform variables (pandas + numpy)

```
import numpy as np

df["lwage"] = np.log(df["wage"])      # logs via numpy
df["exper2"] = df["exper"]**2         # quadratic term
```

- For **dummies/categoricals**, don't recode by hand — let the formula do it with `C(...)`: e.g. `C(sector)` expands a category into dummies (one level kept as baseline).

Estimate and read the output (statsmodels)

```
import statsmodels.formula.api as smf

f = "lwage ~ woman + educ + exper"
m = smf.ols(f, df).fit()
print(m.summary())           # coef, se, t, p-value, 95% CI
```

- Formula syntax: $\sim$ splits y from the regressors; x_1*x_2 adds both terms **and** their interaction; $C(g)$ makes dummies.
- Each summary row is exactly what we covered: $\hat{\beta}$, its se, the t -stat, the p -value, and the 95% CI.

Best practice: robust standard errors by default

```
# reuse the formula f from the previous slide
m = smf.ols(f, df).fit(cov_type="HC3")    # robust SE
m = smf.ols("y ~ x", df).fit(cov_type="HAC",
                             cov_kwds={"maxlags": 4})    # time series
```

- `cov_type="HC3"` leaves $\hat{\beta}$ unchanged and corrects `se`, `t`, `p`, `CI` — make it your **default** in cross-sections.
- For time series use **HAC** (Newey–West); set `maxlags` from the data frequency.

Diagnostics, when you need them (statsmodels.stats)

```
from statsmodels.stats import diagnostic as smd
from statsmodels.stats.stattools import jarque_bera

_, p_white, _, _ = smd.het_white(m.resid, m.model.exog)
_, p_bp, _, _ = smd.het_breuschpagan(m.resid, m.model.exog)
j_b, p_j_b, *_ = jarque_bera(m.resid) # normality
```

- With robust SE already the default, you test mainly to **understand** the data — and to decide whether time-series data need **HAC**. A small p flags the problem.
- **Don't test-then-switch:** pre-testing distorts inference; defaulting to robust SE avoids it. (`het_white` can fail with many regressors — fall back to Breusch–Pagan.)

Reference

The single text for this course — read it for the full, detailed bibliography behind everything in these slides:

Wooldridge, Jeffrey M. 2025. *Introductory Econometrics: A Modern Approach*. 8th ed. Cengage Learning.